

# AP CSP Vocabulary Glossary

Working in Python · alphabetical · 144 terms · AP-tested terms only

Terms marked <sup>AP</sup> appear on the College Board AP CSP exam's own vocabulary list. A superscript chapter number links to where this book defines the term.

Vocabulary drawn from *Working in Python*, a fork of Allen Downey's *Think Python*, 3rd edition (CC BY-NC-SA 4.0), and this course's own AP CSP vocabulary reference. AP CSP definitions here are written in this course's own words, not copied from the College Board's Course and Exam Description.

## A

**abstraction**<sup>AP</sup> Reducing complexity by exposing essential features and hiding implementation detail.

**aggregation of information**<sup>AP</sup> Combining separately harmless pieces of data to identify or profile a person.

**algorithm**<sup>AP</sup> A finite sequence of steps that solves a problem or completes a task. Can be written in English, pseudocode, or code.

**algorithmic efficiency**<sup>AP</sup> An estimate of the computational resources an algorithm uses as a function of input size.

**analog data**<sup>AP 7b</sup> Values that vary continuously and smoothly, with no steps.

**anonymization**<sup>AP</sup> Removing identifying details from data — and its limits, since aggregation can re-identify.

**API**<sup>AP</sup> The specification of how a library's procedures are called and how they behave.

**argument**<sup>AP 2</sup> A value provided to a function when the function is called.

**ASCII**<sup>AP 7b</sup> A table assigning a number from 0 to 127 to each of a small set of characters.

**assignment statement**<sup>AP 2</sup> A statement that assigns a value to a variable.

**authentication**<sup>AP</sup> Verifying that a user is who they claim to be.

## B

**bandwidth**<sup>AP</sup> Maximum data transmittable per unit time, in bits per second.

**beneficial and harmful effects**<sup>AP</sup> Every computing innovation has both, and they can fall on different groups of people.

**bias**<sup>AP</sup> Systematic unfairness embedded in an algorithm or in the data it was built from.

**binary**<sup>AP 7b</sup> Base-2 representation, using only the digits 0 and 1.

**binary search**<sup>AP</sup> Repeatedly halving a sorted list to locate a value.

**bit**<sup>AP 7b</sup> A single binary digit, 0 or 1.

**boolean expression**<sup>AP 5</sup> An expression whose value is either True or False.

**byte**<sup>AP 7b</sup> Eight bits. Enough to hold one of 256 values.

## C

**certificate authority**<sup>AP</sup> An organization that issues and vouches for digital certificates.

**citizen science**<sup>AP</sup> Scientific research carried out partly by volunteers using their own devices.

**code segment**<sup>AP</sup> A portion of a program, one or more lines, treated as a unit.

**collaboration**<sup>AP</sup> Working with others to develop a computing innovation; benefits from varied perspectives and skills.

**comment**<sup>AP 2</sup> Text included in a program that provides information about the program but has no effect on its execution.

**compound conditional**<sup>AP</sup> A single condition combining multiple Boolean expressions with AND, OR, or NOT.

**computer network**<sup>AP</sup> Interconnected computing devices able to send and receive data.

**computer virus**<sup>AP</sup> Malware that copies itself, typically by attaching to other programs or files.

**computing device**<sup>AP</sup> A physical artifact that can run a program.

**computing innovation**<sup>AP</sup> A product, service, or concept that includes a program as an integral part of its function. May be physical, software, or conceptual.

**computing system**<sup>AP</sup> A group of computing devices and programs working together.

**concatenation**<sup>AP 1</sup> Joining two strings end-to-end.

**conditional statement**<sup>AP 5</sup> A statement that controls the flow of execution depending on some condition. (*AP calls this selection.*)

**constant**<sup>AP</sup> A named value that does not change during execution.

**cookies**<sup>AP</sup> Small data files stored by a site to track a user's state or behavior.

**copyright**<sup>AP</sup> The exclusive right of a creator to reproduce, distribute, and display a work.

**correlation**<sup>AP</sup> An association between two variables where changes in one accompany changes in the other.

**creative commons**<sup>AP</sup> A set of public licenses letting a creator pre-authorize specific reuse under stated conditions.

**crowdsourcing** <sup>AP</sup> Gathering input, funding, or labor from many people over the Internet.

## D

**data** <sup>AP</sup> Values that can be stored and processed by a program.

**data abstraction** <sup>AP</sup> Using a data structure, typically a list, to represent related values as one named unit, hiding the individual pieces.

**data cleaning** <sup>AP</sup> Making data uniform and consistent without changing its meaning.

**data mining** <sup>AP</sup> Analyzing large data sets to find patterns and relationships.

**data set** <sup>AP</sup> A collection of related data organized for analysis.

**debugging** <sup>AP 1</sup> The process of finding and correcting errors.

**decidable problem** <sup>AP</sup> A yes/no problem for which an algorithm can always produce the correct answer.

**decimal** <sup>AP 7b</sup> Base-10, the system you already use.

**digital certificate** <sup>AP</sup> A credential issued by an authority binding a public key to an identity.

**digital data** <sup>AP 7b</sup> Values represented in discrete steps, ultimately as bits.

**digital divide** <sup>AP</sup> Unequal access to computing and the Internet across socioeconomic, geographic, or demographic lines.

**distributed computing** <sup>AP</sup> Multiple devices in different locations running parts of one program.

**DNS** <sup>AP</sup> Translates domain names into IP addresses.

**docstring** <sup>AP 4.6b</sup> A string at the beginning of a function that documents what the function does; unlike a comment, it is stored on the function and can be read by the program. (*AP calls this program documentation.*)

## E

**element** <sup>AP 9</sup> One of the values in a list or other sequence.

**encryption** <sup>AP</sup> Encoding data so only holders of the key can read it.

**event** <sup>AP</sup> An action supplied to a program as input: key press, click, sensor reading.

**event-driven programming** <sup>AP</sup> Statements execute in response to events rather than in top-to-bottom sequence.

**expression** <sup>AP 1</sup> A combination of variables, values, and operators.

## F

**fault tolerance** <sup>AP</sup> Continuing to operate when components fail.

**filtering** <sup>AP 10</sup> Looping through a sequence and selecting or omitting elements. (*AP calls this data filtering.*)

**function** <sup>AP 1</sup> A named sequence of statements that performs some useful operation. Functions may or may not take arguments and may or may not produce a result. (*AP calls this a procedure.*)

## H

**hand tracing** <sup>AP 6b</sup> Working through code on paper line by line, writing down each variable's value, in order to find an error without running the program.

**heuristic** <sup>AP</sup> An approach that finds a good-enough solution when finding the optimal one is impractical.

**hexadecimal** <sup>AP 7b</sup> Base-16, using 0 through 9 and A through F. Four bits per digit, so one byte is exactly two hex digits.

**HTTP / HTTPS** <sup>AP</sup> The Web's request/response protocol; HTTPS adds encryption in transit.

## I

**incremental development** <sup>AP 6</sup> A program development plan intended to avoid debugging by adding and testing only a small amount of code at a time.

**index** <sup>AP 8</sup> An integer value used to select an item in a sequence, such as a character in a string. In Python indices start from 0.

**infinite loop** <sup>AP</sup> A loop whose ending condition never becomes true.

**information** <sup>AP</sup> Meaning, patterns, or insight extracted from data.

**integer** <sup>AP 1</sup> A type that represents numbers with no fractional or decimal part.

**intellectual property** <sup>AP</sup> Creative or technical work legally owned by its creator.

**interface design** <sup>AP 4</sup> A process for designing the interface of a function, which includes the parameters it should take. (*AP calls this procedural abstraction.*)

**internet** <sup>AP</sup> A network of interconnected networks using standardized open protocols.

**IP** <sup>AP</sup> Protocol for addressing devices and routing packets between them.

**IP address** <sup>AP</sup> The unique identifier assigned to a device on a network.

**iterative development** <sup>AP</sup> Repeated cycles of build, feedback, and revision; earlier phases get revisited.

## K

**keylogging** <sup>AP</sup> Recording keystrokes to capture passwords and confidential input.

## L

**linear search** <sup>AP 7</sup> A computational pattern that searches through a sequence of elements and stops when it finds what it is looking for.

**list** <sup>AP 9</sup> An object that contains a sequence of values. (*on the AP exam, list indices start at 1; in Python, at 0.*)

**logical operator**<sup>AP 5</sup> One of the operators that combines boolean expressions, including and, or, and not.

**loop**<sup>AP 3</sup> A statement that runs one or more statements, often repeatedly. (*AP calls this iteration.*)

**lossless compression**<sup>AP 7b</sup> Reduces size while allowing the original to be reconstructed exactly.

**lossy compression**<sup>AP 7b</sup> Reduces size further, but only an approximation of the original can be recovered.

## M

**machine learning**<sup>AP</sup> Systems that improve at a task from data rather than from explicit programming.

**malware**<sup>AP</sup> Software intended to damage or take control of a system.

**metadata**<sup>AP</sup> Data describing other data: creation date, size, author, location, format.

**modularity**<sup>AP</sup> Dividing a program into independent parts each responsible for one aspect.

**module**<sup>AP 2</sup> A file that contains Python code, including function definitions and sometimes other statements. (*AP calls this a software library.*)

**modulus operator**<sup>AP 5</sup> An operator, %, that works on integers and returns the remainder when one number is divided by another. (*AP calls this MOD.*)

**multifactor authentication**<sup>AP</sup> Requiring evidence from two or more categories: knowledge, possession, inherence.

## N

**nested conditional**<sup>AP 5</sup> A conditional statement that appears in one of the branches of another conditional statement.

## O

**open access**<sup>AP</sup> Research or data made available free of access restrictions.

**open source**<sup>AP</sup> Software whose source is public and may be modified and redistributed.

**open standard**<sup>AP</sup> A publicly available specification that anyone may implement without permission or fee.

**overflow error**<sup>AP 7b</sup> An error that happens when a value is too large for the number of bits available to hold it.

## P

**packet**<sup>AP</sup> A chunk of data carrying metadata such as its destination, sent across a network.

**packet switching**<sup>AP</sup> Splitting a message into packets that travel independently, possibly by different routes, and are reassembled at the destination.

**pair programming**<sup>AP</sup> Two programmers at one workstation: one drives, one reviews and thinks ahead.

**parallel computing**<sup>AP</sup> A program split into operations, some performed simultaneously.

**parameter**<sup>AP 3</sup> A name used inside a function to refer to the value passed as an argument.

**phishing**<sup>AP</sup> Tricking a user into revealing private information by impersonating a trusted party.

**PII**<sup>AP</sup> Information that identifies or can be linked to a specific individual.

**pixel**<sup>AP</sup> The smallest addressable color element of a display or image.

**plagiarism**<sup>AP</sup> Presenting someone else's work as your own.

**procedure**<sup>AP 6b</sup> The exam's word for a named, reusable block of code, whether or not it returns a value. This book says function. In older languages the two words were distinct: a procedure returned nothing, a function returned a value.

**program**<sup>AP</sup> A collection of statements that performs a task when run.

**program function**<sup>AP 6b</sup> What a program does when it runs, described as behavior.

**program input**<sup>AP 6b</sup> Data a program receives while it is running.

**program output**<sup>AP 6b</sup> What a program produces: displayed text, a returned value, a file, a sound, a movement.

**program purpose**<sup>AP 6b</sup> The need a program serves, or the problem it solves; why it exists.

**protocol**<sup>AP</sup> An agreed set of rules governing system behavior.

**pseudorandom**<sup>AP 12</sup> A pseudorandom sequence of numbers appears to be random, but is generated by a deterministic program. (*AP calls this RANDOM(a, b) — inclusive on both ends, unlike Python's range.*)

**public key encryption**<sup>AP</sup> A public key encrypts; a different private key decrypts.

## R

**reasonable time**<sup>AP</sup> Run time that grows polynomially or slower with input size.

**redundancy**<sup>AP</sup> Duplicate paths or components that allow operation to continue after a failure.

**relational operator**<sup>AP 5</sup> One of the operators that compares its operands: ==, !=, >, <, >=, and <=.

**return value**<sup>AP 6</sup> The result of a function. If a function call is used as an expression, the return value is the value of the expression. (*AP calls this the RETURN statement.*)

**RGB**<sup>AP 7b</sup> Color stored as three numbers, the amounts of red, green, and blue, each usually one byte.

**rogue access point**<sup>AP</sup> An unauthorized wireless access point used to intercept network traffic.

**roundoff error**<sup>AP 6b, 7b</sup> A loss of precision that happens because a fixed number of bits cannot represent some numbers exactly.

**router**<sup>AP</sup> A device that forwards packets between networks toward their destination.

**routing**<sup>AP</sup> Determining a path from sender to receiver across a network.

**runtime error**<sup>AP 2</sup> An error that causes a program to display an error message and exit.

## S

**sampling**<sup>AP 7b</sup> Approximating an analog signal by measuring it at regular intervals. Two independent settings: how often you measure (rate) and how precisely you record each measurement (bit depth).

**scalability**<sup>AP</sup> A system's ability to keep working acceptably as demand grows.

**semantic error**<sup>AP 2,6b</sup> An error that lets the program run but produces a wrong result. (*AP calls this a logic error.*)

**sequencing**<sup>AP</sup> Executing steps in the order written.

**sequential computing**<sup>AP</sup> Operations performed one at a time, in order.

**simulation**<sup>AP</sup> An abstraction of a real phenomenon using varying values to represent changing states, run for a purpose.

**speedup**<sup>AP</sup> Sequential time divided by parallel time.

**string**<sup>AP 1</sup> A type that represents sequences of characters.

**substring**<sup>AP</sup> A contiguous portion of a string.

**symmetric key encryption**<sup>AP</sup> One shared key both encrypts and decrypts.

**syntax error**<sup>AP 1</sup> An error in a program that makes it impossible to parse — and therefore impossible to run.

## T

**TCP**<sup>AP</sup> Transport protocol providing reliable, ordered delivery with retransmission.

**test case**<sup>AP 6b</sup> A specific input paired with its expected output. Good sets include typical, boundary, and edge values.

**testing**<sup>AP 6b</sup> Checking that a program behaves correctly by running it on chosen inputs and comparing what comes out against what should have come out.

**traversal**<sup>AP</sup> Iterating over the items of a list. Full traversal visits every element; partial stops early.

**type**<sup>AP 1</sup> A category of values. The types we have seen so far are integers (type int), floating-point numbers (type float), and strings (type str). (*AP calls this data type.*)

## U

**UDP**<sup>AP</sup> Lightweight transport protocol with minimal error checking; faster, less reliable.

**undecidable problem**<sup>AP</sup> A problem for which no algorithm can always give a correct yes/no answer for every input.

**Unicode**<sup>AP 7b</sup> A far larger table, covering the writing systems ASCII left out.

**unreasonable time**<sup>AP</sup> Run time that grows exponentially or factorially with input size.

## V

**variable**<sup>AP 2</sup> A name that refers to a value.

## W

**world wide web**<sup>AP</sup> A system of linked pages and files that runs on top of the Internet using HTTP.